DOC: Improving Long Story Coherence With Detailed Outline Control

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Abstract

We propose the Detailed Outline Control (DOC) framework for improving long-range plot coherence when automatically generating several-thousand-word-long stories. DOC consists of two complementary components: a detailed outliner and a detailed controller. The detailed outliner creates a more detailed, hierarchically structured outline, shifting creative burden from the main drafting procedure to the planning stage. The detailed controller ensures the more detailed outline is still respected during generation by controlling story passages to align with outline details. In human evaluations of automatically generated stories, DOC substantially outperforms a strong Re^3 baseline (Yang et al., 2022) on plot coherence (22.5% absolute gain), outline relevance (28.2%), and interestingness (20.7%). Humans also judged DOC to be much more controllable in an interactive generation setting.

1 Introduction

Recent advancements in natural language generation systems have fueled increased interest in *long-form* text generation, in which texts may span thousands of words or more. Compared to tasks with shorter outputs, long-form generation involves meaningfully different challenges. It is nontrivial to maintain overarching coherence, or even basic relevance to an initial premise or plan. Even the most advanced language models to date, such as GPT4 (OpenAI, 2023), still cite long context as a major direction for further improvement, and require structured planning to generate text longer than a few hundred words.

In this work, we focus on long-form *story* generation, which is representative of the major difficulties in long text generation. Only recently have prior efforts even attempted to generate stories of comparable length to human-authored "short stories" (Re^3, Yang et al. (2022)). Compared to humans, state-of-the-art story generation systems like

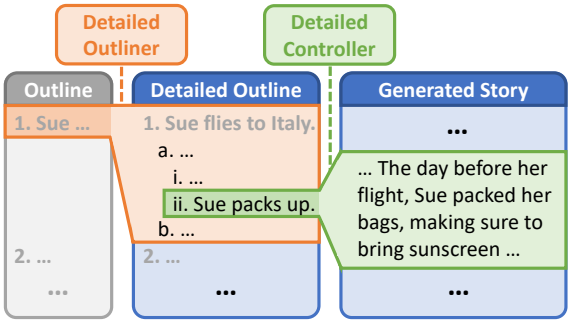

Figure 1: High-level overview of DOC. Our detailed outliner expands a brief initial outline into a more detailed outline. Our detailed controller then maintains faithfulness to the more detailed outline when drafting the main story.

Re^3 still fall short in numerous areas: common failure modes include insufficient high-level planning resulting in local fluency amid global incoherence, or deviating from said planning even when it exists.

To bridge some of this gap, we propose the Detailed Outline Control (DOC) framework. While reusing the high-level planning-drafting-revision structure of Re^3, DOC improves long-range plot coherence via two complementary approaches.

First, our *detailed outliner* refines a brief initial outline into a more detailed, hierarchical one (Figure 1 left). As motivation, a human author might also iteratively refine and expand a brief initial outline before drafting a long document, using the outline to guide a coherent plot rather than improvising plot points on the fly. Accordingly, our detailed outliner employs a structured prompting procedure to create a detailed outline with length scalable according to the desired scope of generation. Individual outline items are associated with a setting and characters, and are carefully filtered for relevance and coherence in context.

Second, our *detailed controller* maintains faithfulness to our detailed outline by controlling passage generation based on corresponding outline items (Figure 1 right). Because our detailed outline imposes many overlapping soft constraints, the detailed controller must exert sufficient control

strength to enforce them. The detailed controller must also accommodate flexible natural language inputs and be computationally efficient when generating with state-of-the-art large language models. We implement the detailed controller as an OPT-350m-based controller according to FUDGE (Yang and Klein, 2021), designing a contrastive training procedure that aligns summaries to passage prefixes. In particular, we construct fluent hard negatives to encourage lengthy outputs to be not only initially on topic, but relevant throughout.

Compared to the original Re^3, the previous state of the art in long-form story generation, using DOC achieves dramatically higher plot coherence (22.5% absolute gain), outline relevance (28.2%), and even interestingness (20.7%) in pairwise human evaluations (Section 4). Our ablations indicate that both the detailed outliner and detailed controller are critical (Section 5.1). We also demonstrate that DOC can generate stories in collaboration with humans, interacting at a high-level planning stage rather than passage-by-passage as in many prior works (Coenen et al., 2021; Lee et al., 2022), and is overwhelmingly preferred over the original Re^3 in this setting (Section 4.1).[1]

2 Related Work

Although we generate stories an order of magnitude longer compared to most prior works (Wang and Wan, 2019; Yao et al., 2019; Qin et al., 2019; Xu et al., 2020; Wang et al., 2022), we highlight below several works which employ related ideas.

Hierarchical Generation. A key component of DOC is our detailed outliner, which generates an outline hierarchically. Hierarchical structure in long-form generation can be implemented as part of the model architecture itself (Yang et al., 2016; Miculicich et al., 2018; Guo et al., 2021), or as natural language outlines or structured schema (Fan et al., 2018; Yao et al., 2019; Goldfarb-Tarrant et al., 2020; Rashkin et al., 2020; Zhao et al., 2020; Narayan et al., 2021; Tian and Peng, 2022; Mirowski et al., 2022; Yang et al., 2022). DOC's detailed outliner also builds a natural language outline, but can easily increase the level of detail to match the desired scope of the final story.

Controlled Generation. A second key component of DOC is the detailed controller, which increases faithfulness to our detailed outline. Prior works such as Hu et al. (2019) use constrained decoding to guarantee rule-based constraints, while Dathathri et al. (2019); Krause et al. (2020); Yang and Klein (2021) propose modular control schemes based on an auxiliary model for a desired attribute. However, such methods typically do not handle natural language instructions.

In contrast, prompting (Brown et al., 2020; Zhong et al., 2021; Sanh et al., 2021; Wu et al., 2022; Kojima et al., 2022; Ouyang et al., 2022) offers a lightweight, flexible alternative. However, while prompts are an effective way to *provide context*, they may be insufficient for *enforcing constraints* due to the limited control strength, which is not easily tunable unlike in our detailed controller.

Human-In-The-Loop Story Generation. Some previous works generate longer stories with a human in the loop (Goldfarb-Tarrant et al., 2019; Coenen et al., 2021; Lee et al., 2022; Chung et al., 2022; Ippolito et al., 2022; Mirowski et al., 2022). We emphasize that DOC is designed to generate stories without human intervention. Nevertheless, due to planning in natural language space, DOC is in principle highly human-controllable. Unlike methods which interact with the human passage by passage (Coenen et al., 2021; Lee et al., 2022), DOC can also interact at a higher-level planning stage, as explored in Section 4.1.

3 Detailed Outline Control

We introduce the Detailed Outline Control (DOC) framework, aiming to improve long-range plot coherence in automatically generated long stories.

3.1 Background and Motivation

A major inspiration for our work is Re^3 (Yang et al., 2022), which generates plot-coherent long-form stories of over 2000 words by decomposing the writing process into planning, drafting, rewriting, and editing steps. Their high-level plan contains a setting, character inventory, and brief three-point outline (e.g., Figure 1 "Outline"). In particular, when drafting each next story passage, they inject relevant context from the high-level plan and previously generated story via structured prompting (Figure 2). They finally rerank possible continuations using rerankers for outline relevance and passage coherence, and edit for consistency. DOC follows the high-level writing process and structured-prompting-based passage generation proposed by

[1] All code and models are available at https://github.com/yangkevin2/doc-story-generation.

Structured Prompt For Drafting	
Characters	1. Daisy is a kind-hearted old woman. She has cancer. Bill is her husband. 2. Lisa is Daisy's daughter.
Summary (Far Past)	Daisy is diagnosed with cancer. Lisa is trying to find a viable treatment.
Summary (Near Past)	Lisa has been stressed out lately, and Daisy expresses her concern.
Upcoming Events	Lisa tirelessly continues her research. Lisa finally finds a cure. Setting: Lisa's laboratory.
Verbatim Preceding Passage	Lisa looked back at Daisy, her eyes clear and full of determination. "I've got this, mom. Hang in there."

Figure 2: Stylized example showing the main components of the structured prompt used to draft new story passages in Re^3 and DOC. Leveraging our detailed outline and detailed controller, new elements of DOC's prompt include character development over time (red), more detailed events based on outline leaf nodes (orange), future context (green), and improved setting and character information (purple).

Yang et al. (2022), though we remove the time-consuming editing step, which they find does not significantly affect final story quality.

However, Yang et al. (2022) note that despite greatly outperforming simple rolling-window baselines, Re^3 still makes frequent errors in long-range coherence: some stories still contain lengthy passages which seem not to fit the surrounding context, or deviate heavily from the initial outline. DOC aims to address these shortcomings via two major innovations: more detailed planning via our detailed outliner, and correspondingly fine-grained control during drafting via our detailed controller.

Detailed Outliner Motivation. While Re^3's outlines are plausible, they are insufficiently concrete, and do not scale to longer stories. A human author would not write a novel given just a three-sentence beginning, middle, and end. Not only can a more detailed outline empirically result in improved plot coherence (Section 4), but it can enable greater control in human interaction as well (Section 4.1).

Therefore, DOC constructs a detailed outline (e.g., Figure 1 "Detailed Outline") with depth adjustable according to the desired length of the final story. The detailed outline shifts creative burden from drafting to planning, reducing the need to improvise plot points on the fly during drafting.

Detailed Controller Motivation. The greater level of detail in our outline makes it much harder to stay faithful to that outline. To work with large language models such as GPT3-175B during drafting, prior works such as Re^3 have relied on clever prompting together with rejection sampling or reranking. However, prompting and reranking approaches are limited in the strength of control they can exert over the model distribution, which is especially problematic for systems like Re^3 which rely on complex constraints and long context in a structured prompt. Indeed, Yang et al. (2022) observe that many of Re^3's stories already omit parts of even their brief three-point outline—and DOC's outline will impose far more detailed constraints.

Therefore, we design DOC's detailed controller to more strongly enforce the complex natural language constraints set by the outline. Our detailed controller, an adaptation of FUDGE (Yang and Klein, 2021), will operate token-by-token throughout generation instead of relying on only an initial prompt or post-hoc rejection sampling.

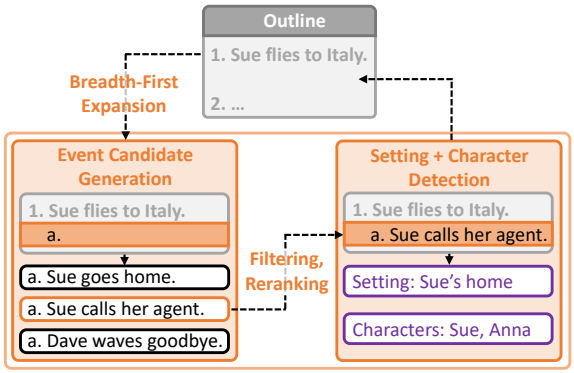

Figure 3: Diagram of new entry creation in the detailed outline. Our detailed outliner recursively expands outline items in breadth-first order. To create each new entry, it proposes candidate events and selects the best via filtering and reranking, and then detects the setting and relevant characters.

3.2 Detailed Outliner

Our detailed outliner recursively generates a hierarchical detailed outline at arbitrary granularity. Figure 3 summarizes the individual components.

Breadth-First Expansion. Viewing the outline as a tree $\mathbf{T}$ initialized as just a root node r, we generate children in breadth-first expansion order. Starting from the items of the initial top-level outline (depth 1), we generate all of their children (depth 2), then all childrens' children (depth 3), and so forth. For each parent node p, we generate children one by one, stopping when a child c's event description ends with the end-of-text token. We restart and resample for a given p if there are too few or too many children, although empirically this procedure almost always results in just two or three children. We terminate outline generation after reaching a pre-specified depth.